Rossmoyne Senior High School

Semester One Examination, 2018

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One: Calculator-free

Your first name ____

Your surname ____

Circle your teacher's name: Mr Fletcher Mr Pisano

Time allowed for this section

Reading time before commencing work:	five minutes
Working time:	fifty minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet

To be provided by the candidate

Standard items:	pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters
Special items:	nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet.

3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.

4. Supplementary pages for the use of planning/continuing your answer to a question have been provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

6. It is recommended that you do not use pencil, except in diagrams.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free **35% (52 Marks)**

This section has **seven (7)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1 **(5 marks)**

The rate of the Goods and Services Tax (GST) varies from country to country. In Australia it is 10% but in Thailand it is only 7% whilst in Sweden it is 25%.

(a) Determine the amount of GST that must be added to goods valued at $300 in Sweden. (1 mark)

(b) The price of a TV set in Thailand is $400, excluding GST. Calculate the GST inclusive price. (2 marks)

(c) In another country, a $20 CD costs $23 when GST is included. Determine the rate of GST in this country. (2 marks)

Question 2 **(9 marks)**

A phone repair business employs casual labourers at busy times. The business pays casual labourers an hourly rate of \$30 per hour, with time and a half being paid for any weekend work.

One week, a casual labourer worked 4 hours on Monday, 3 hours on Tuesday, 3 hours on Friday, 3 hours on Saturday and 2 hours on Sunday.

(a) How much did the casual labourer earn

(i) on Friday? (1 mark)

(ii) over the weekend? (2 marks)

(iii) during the whole week? (2 marks)

(b) Calculate the **increase** in earnings the following week, if the casual labourer worked the same days but for 3 hours each day. (2 marks)

(c) The business makes a superannuation contribution of 12.5% of weekly earnings for casual labourers. Calculate the weekly superannuation contribution for a casual labourer who earns \$824 per week. (2 marks)

Question 3 **(8 marks)**

The spreadsheet below shows some example costs (C in \$) to transport a package by air between two locations. The cost depends on the weight of the package (W kg) and the amount by which the longest dimension of the package exceeds one metre (E m).

C (\$)		E (m)			
		0	1	2	3
W (kg)	10	30	42	58	78
	20	60	82	108	138
	30	90	122	A	198
	40	120	162	208	258

(a) Use the spreadsheet to determine C when

(i) $E = 1$ and $W = 40$. (1 mark)

(ii) $W = 10$ and $E = 3$. (1 mark)

The cost is calculated using the formula $C = 3W + 2E^2 + EW$.

(b) Show use of the above formula to determine the value of A in the spreadsheet. (3 marks)

(c) Calculate the cost of transporting a 15 kg package that has a longest dimension of 3 m. (3 marks)

Question 4 **(8 marks)**

Four measurements are recorded on a production line at hourly intervals and then used to calculate the values of P_1, P_2 and P_3. These values are then reported to a supervisor. The formulas used are shown below.

$$P_1 = d^2 - \frac{1}{2}c^2, \qquad P_2 = \frac{40}{a} + b\sqrt{d}, \qquad P_3 = 100P_2 + 10P_1.$$

During one round of measurements, the values recorded were

$$a = 8, \qquad b = 11, \qquad c = 6, \qquad d = 9.$$

Calculate the value of

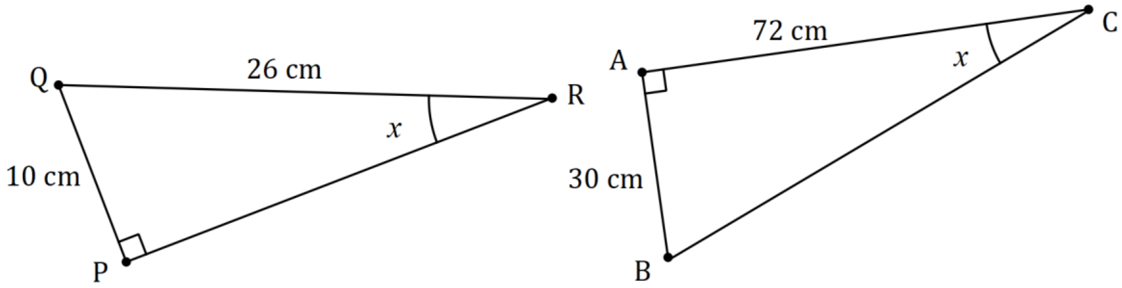
(a) P_1. (3 marks)

(b) P_2. (3 marks)

(c) P_3. (2 marks)

Question 5 **(7 marks)**

(a) Two similar right-triangles (not drawn to scale) are shown below.



(i) Determine the lengths of the sides BC and PR. (3 marks)

(ii) The area of $\triangle ABC$ is k times greater than the area of $\triangle PQR$. State the value of k. (1 mark)

(b) A right-triangle has sides of lengths 9 cm, 41 cm and 40 cm.

(i) Between which two sides is the right-angle? (1 mark)

(ii) Calculate the area of the triangle. (2 marks)

Question 6 **(5 marks)**

Summary information about a small portfolio of shares is shown in the table below.

Share code	Number of shares	Current market price	Last dividend per share	Price-Earnings ratio
XBC	500	\$4.50	\$0.08	17.5
CFW	800	\$15.00	\$0.60	14.0

(a) Determine the total dividend of the portfolio, based on the last dividend paid. (2 marks)

(b) The investor sells all the CFW shares at the current market price, paying a 3% commission for brokerage. Determine how much the investor receives. (3 marks)

Question 7 **(10 marks)**

(a) Express $4\begin{bmatrix}2 & -10 & 5\end{bmatrix} + 2\begin{bmatrix}-2 & -4 & 5\end{bmatrix}$ as a single matrix. (3 marks)

(b) Calculate $\begin{bmatrix}5 & 0 & 1\\0 & -2 & -4\end{bmatrix} \times \begin{bmatrix}2\\-1\\3\end{bmatrix}$. (2 marks)

(c) Matrices P and Q, with dimensions 4×6 and 7×4 respectively, can be multiplied together. State the dimensions of the product, briefly justifying your answer. (2 marks)

(d) Determine the values of the constants a, b and c given that $5I + a\begin{bmatrix}-1 & b\\2 & -4\end{bmatrix} = \begin{bmatrix}3 & 12\\4 & c\end{bmatrix}$, where I is the 2×2 identity matrix. (3 marks)

End of questions

Supplementary page

Question number: _________

Supplementary page

Question number: _________